DI-04 CHIMNEY CENSUS

Orbital Brick Kiln Detection & Tactical Command System | Complete Technical & Architecture Manual

1. Executive Mission & Problem Formulation

Under 2022 environmental directives across South Asia, traditional high-polluting **Fixed Chimney Bull's Trench Kilns (FCBK)** are legally mandated to convert to cleaner **Zigzag** firing technology. With 1 inspection jeep and 2 field officers covering 3,000 km², field surveillance is physically impossible. European Space Agency **Sentinel-2 L2A satellite imagery** covers the entire district at 10 m/pixel resolution. In orbit, a kiln spans merely **12 × 6 pixels**. The DI-04 Chimney Census system converts raw orbital imagery into oriented bounding boxes, distinct physical facility censuses, and ranked Top 20 inspection field directives.

2. System Architecture & End-to-End Pipeline

STAGE 1: DATA PIPELINE	STAGE 2: NEURAL DETECTION	STAGE 3: GEOSPATIAL INTELLIGENCE	STAGE 4: ENFORCEMENT & DEPLOYMENT
• SentinelKilnDB Loader • Quality & Balance Audit • YOLO-OBB Normalizer • 128×128 Tile Slicing	• YOLOv8n-OBB Architecture • 3.08M Params / 8.44 GFLOPs • Rotated CIoU + DFL Loss • Sub-2.2ms GPU Inference	• WGS-84 Coordinate Transform • 30px Tile Deduplication • Wilson Uncertainty Intervals • 1 km Route Cluster Analysis	• Inspector's Brief (2-Page PDF) • Top 20 Inspection Targets • Streamlit Command Center • Raspberry Pi 5 Edge Engine

3. Software Tools, Frameworks & Why We Use Them

Tool / Library	Project Role	Engineering Rationale & Benefit
PyTorch 2.2+	Deep Learning Backbone	Hardware-accelerated CUDA execution with Automated Mixed Precision (AMP) for rapid training.
Ultralytics YOLO-OBB	Rotated Object Detector	Natively predicts 8 corner coordinates [x1..y4] + theta, eliminating 70% background agricultural clutter.
Shapely	Geometric Computation	Calculates exact polygon intersection-over-union and repairs polygon self-intersections with buffer(0).
ONNX Runtime	Edge Hardware Engine	Fuses neural layers for ultra-lightweight CPU execution (2.96 ms/tile, 337 FPS) on Raspberry Pi 5.
OpenCV (cv2)	Vision & Pushbroom Stream	Fast video streaming, tile slicing, HUD overlays, and temporal Earth-space frame tracking.
Streamlit	Command Center Platform	Renders 6 interactive mission workspaces with zero frontend latency and custom dark aerospace CSS.
ReportLab	PDF Brief Generation	Builds field-ready, strictly 2-page operational inspection directives with exact point typography.
Pandas & NumPy	Geospatial & Vector Math	Haversine geodesic distance, Wilson confidence bounds, and multi-criteria prioritization formulas.
Pytest	Automated Testing Suite	Guarantees code reliability with 24 passing unit tests covering geometry, schemas, and runtime limits.

4. Complete Directory Layout & Function-by-Function Catalog

Every module in the codebase is modular, tested, and mapped to a specific stage in the mission pipeline:

File Path	Core Functionality	Key Functions & Purpose
run_pipeline.py	Master Pipeline CLI	main(): Parses CLI modes (smoke, audit, prepare, train, eval, predict, report, all). run_command_stage(): Subprocess execution and logging.
app.py	Command Center Platform	load_yolo_model(): Caches YOLO-OBB checkpoint. compute_obb_metrics(): Derives physical area (m²), orientation angle (0-180°), width/height. Implements 6 interactive mission workspaces.
src/data/loader.py	Hugging Face Data Ingestion	download_sentinel_kilndb(): Fetches SentinelKilnDB tiles from Hugging Face. load_split(): Loads train, val, test splits into memory. get_tile_by_id(): Retrieves tile imagery & annotations.
src/data/audit.py	Dataset Quality Audit	audit_image_dimensions(): Validates 128x128x3 tile dimensions. audit_class_distribution(): Quantifies CFCBK, FCBK, Zigzag counts. audit_coordinate_validity(): Detects degenerate or inverted boxes. generate_audit_report(): Exports structured audit JSON.
src/data/yolo_dataset.py	YOLO-OBB Format Exporter	convert_polygon_to_yolo_obb(): Normalizes 8 coordinates to [0,1]. export_yolo_dataset(): Writes image & label txt pairs for Ultralytics. generate_contact_sheet(): Renders visual contact sheet.
src/models/train.py	YOLO-OBB Model Training	configure_training(): Sets AdamW, cosine LR, warmup, and loss weights. train_yolo_obb(): Executes mixed-precision training; exports best.pt checkpoint.
src/models/predict.py	Batched Inference Engine	run_batched_inference(): High-throughput forward passes with Rotated NMS. format_submission_row(): Formats 8-coord CSV submission rows. generate_submission_csv(): Exports competition submission CSV.
src/evaluation/rotated_iou.py	Polygon Geometry & IoU	polygon_from_8coords(): Builds Shapely Polygon with buffer(0) fix. calculate_rotated_iou(): Computes exact intersection over union.
src/evaluation/dota_evaluator.py	Official DOTA Evaluator	compute_ap_all_points(): All-point PR curve interpolation AP. evaluate_predictions(): Computes per-class and mean rotated mAP@0.5.
src/geospatial/transform.py	Geographic Coordinate Projection	tile_pixel_to_wgs84(): Converts pixel offsets to WGS-84 Lat/Lon. export_to_geojson(): Exports RFC 7946 compliant GeoJSON feature layer.
src/geospatial/deduplication.py	Overlapping Tile Deduplicator	calculate_haversine_distance(): Haversine surface distance. deduplicate_district_detections(): Merges 30px tile overlap detections (<=50m).
src/geospatial/analysis.py	Uncertainty & Proximity Analysis	calculate_uncertainty_intervals(): Wilson score confidence intervals. find_nearby_kilns(): Identifies route cluster pairs within 1 km.
src/geospatial/priority.py	Multi-Criteria Field Ranking	score_inspection_candidate(): 0.35*Type + 0.30*Density + 0.20*Hist + 0.15*Conf. generate_top20_priority_list(): Exports Farida Khan's Top 20 targets CSV.
src/reporting/map_generator.py	District Cartographic Map	render_district_map(): Plots kilns, cluster links, scale bar, and north arrow.
src/reporting/pdf_brief.py	Inspector's Brief Generator	generate_inspectors_brief_pdf(): Assembles 2-page field inspection PDF.
src/reporting/presentation.py	Presentation & Script Generator	create_briefing_deck(): Exports 5-slide PPTX deck. create_speaker_notes(): Writes 5-minute timed speaker script.

deploy/raspberry_pi/ raspberry_pi_satellite_edge.py	Autonomous Edge AI Engine	EarthSpaceTracker: Tracks stationary ground targets across frames. apply_frame_nms(): Cross-frame oriented NMS. run_edge_inference(): Real-time pushbroom stream processing & downlink telemetry.

5. Core Mathematical Logic & Algorithmic Formulation

A. Oriented Bounding Box (OBB) Corner Representation

Instead of axis-aligned boxes (x, y, w, h) that enclose unnecessary background, each kiln is represented by 4 ordered vertices in normalized tile coordinates:
Box = [(x■, y■), (x■, y■), (x■, y■), (x■, y■)] where x■, y■ ∈ [0.0, 1.0]. This allows rotation angle θ ∈ [0°, 180°) to tightly hug the kiln perimeter and completely eliminate false overlap with adjacent kilns.

B. Multi-Task Training Loss Formulation

The model is optimized using a weighted multi-task objective:
Loss = 7.5 · $L_{Rotated\text{-}CIoU}$ + 0.5 · L_{BCE} + 1.5 · L_{DFL}
• $L_{Rotated\text{-}CIoU}$: Penalizes polygon overlap, distance between centroids, and aspect ratio discrepancy.
• L_{BCE}: Binary Cross-Entropy with class balancing weights to protect rare CFCBK (3%) from starvation.
• L_{DFL}: Distribution Focal Loss optimizes sub-pixel corner coordinate regression around 12×6 pixel targets.

C. Spatial Deduplication Across Overlapping Tiles

Adjacent satellite tiles have a 30-pixel overlap (~300 meters). Two detections A and B from adjacent tiles are merged if:
$Distance_{Haversine}(Centroid_A, Centroid_B) \leq 50.0$ meters AND Rotated-IoU(A, B) ≥ 0.30
This guarantees zero double-counting on the district map while maintaining raw tile predictions for leaderboard submission.

D. Multi-Criteria Field Inspection Ranking

Inspector Farida Khan's priority score is calculated using four operational criteria:
Priority = 0.35 · Type + 0.30 · Cluster Density + 0.20 · Historical Status + 0.15 · Model Confidence
• **Type Scrutiny (0.35)**: FCBK = 1.0 (Non-compliant), CFCBK = 0.9, Zigzag = 0.2 (Compliant).
• **Cluster Density (0.30)**: Kilns within 1 km radius normalized to [0,1], maximizing inspection route efficiency.
• **Historical Status (0.20)**: Persisting unconverted FCBKs from 2023-2024 records receive highest urgency.
• **Confidence (0.15)**: Continuous model confidence score, filtering out borderline ambiguous activations.

6. Measured System Benchmarks & Performance Profile

Evaluation Metric / Property	Measured Result	Operational Implication
Validation DOTA mAP@0.5	**0.782**	Rotated polygon benchmark across all 3 classes.
Zigzag Clean Precision / Recall	**0.85 / 0.81 (F1 = 0.83)**	High precision confirms genuine clean technology conversions.
FCBK Legacy Precision / Recall	**0.82 / 0.79 (F1 = 0.80)**	High recall catches non-compliant polluters across rural zones.
GPU Inference Latency	**2.2 ms / tile** (454 FPS)	8,000 test tiles evaluated in < 20 seconds (Limit: 15 min).
Edge CPU Latency (RPi 5)	**2.96 ms / tile** (337 FPS)	Lightweight ONNX runtime runs easily within 4.2W CubeSat power budget.
Total Model Parameters	**3,083,100** (8.44 GFLOPs)	Ultra-compact 6.06 MB state dict, 100% trainable.
Automated Unit Tests	**24 / 24 Tests Passing**	100% verified across geometry, schemas, and edge runtimes.